

# Certificate of Analysis

## PRODUCT DETAILS

**Name**

**Product Name**

General Exosomes (RejuvEXO)

**Batch**

**Manufacturing Date**

**Expiry Date**

**Dosage Form**

Clear-colourless solution

**Storage Temperature**

4 – 8°C

| TEST                                 | SPECIFICATION                                                 | TEST RESULTS                      | REMARKS                                        |
|--------------------------------------|---------------------------------------------------------------|-----------------------------------|------------------------------------------------|
| Infectious Diseases Screening        | HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR                      | Not Detected / Non-Reactive       | <div>QUALITY CONTROL</div> <div>APPROVED</div> |
| Sterility                            | No cloudiness of the test media                               | Negative                          | <div>QUALITY CONTROL</div> <div>APPROVED</div> |
| Bacterial Endotoxin Test (LAL)       | Endotoxin sensitivity 0.125 EU/mL                             | Negative                          | <div>QUALITY CONTROL</div> <div>APPROVED</div> |
| Mycoplasma                           | Mean of absorbance <0.05                                      | Negative                          | <div>QUALITY CONTROL</div> <div>APPROVED</div> |
| Simple Western Blot                  | Positive Marker:<br>Tetraspanin – CD9, CD63                   | Detected                          | <div>QUALITY CONTROL</div> <div>APPROVED</div> |
|                                      | Cytosolic – TSG101<br>Negative Marker:<br>Cellular – Calnexin | Trace marker detected             | <div>QUALITY CONTROL</div> <div>APPROVED</div> |
| Nanoparticle Tracking Analysis (NTA) | Exosome particle size range 30nm – 200nm                      | 34 x 10 <sup>9</sup> particles/mL |                                                |

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Storage recommendation:

- 1) Refrigerate the vials in 4-8°C at all times upon received
- 2) Keep away the vials from direct heat or sunlight
- 3) Avoid prolong exposure in room condition
- 4) It is recommended to place the vials in freezer for prolonged storage

**Authorised Signature:**

**Quality Control Department**

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## Supporting Information

### STERILITY TEST <sup>6, 8</sup>

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.  
Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

### BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST <sup>3,6</sup>

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

### MYCOPLASMA TEST <sup>4,6</sup>

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

### SIMPLE WESTERN BLOT

| Marker | Positive Marker |      | Negative Marker |          |
|--------|-----------------|------|-----------------|----------|
|        | Tetraspanin     |      | Cytosolic       | Cellular |
|        | CD9             | CD63 | TSG101          | Calnexin |
| Result | +               | +    | +               | *        |

'+' indicates the presence of the marker in the cell.  
'\*' indicates the trace of the marker detected in the cell

### NANOPARTICLE TRACKING ANALYSIS (NTA) <sup>7,8</sup>

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.



#### REFERENCES

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- Joana Maia, Silvia Batista, Nuno Couto, Ana C. Gregório, Cristian Bodo, Julia Elzanowska, Maria Carolina Strano Moraes, Bruno Costa-Silva, 2020, Employing Flow Cytometry to Extracellular Vesicles Sample Microvolume Analysis and Quality Control.
- Bacterial Endotoxins. European Pharmacopoeia 5.0; <https://empoa.com/validation/Method/LAL/EUPHARMACPOE05A.pdf>
- US Pharmacopoeia <63> Mycoplasma Tests
- ICH Harmonised Tripartite Guideline. Specifications: Test Procedures and Acceptance Criteria for Biotechnological/ Biological Products Q5B.
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- Neha Chopra, Ibrahim Dutt Arya, Namrata Jain, Poonam Yadav, Salma Wajid, Sarinder P. Singh, Sangeta Choudhary, 2019, Biophysical Characterization and Drug Delivery Potential of Exosomes from Human Wharton's Jelly-Derived Mesenchymal Stem Cells.
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